

My AoPs Solution Intermediate Algebra

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1 Substitution Exercises

1. Find all ordered pairs (x, y) such that $-x + 2y = 8$ and $x - y = -5$

$$y = x + 5$$

Substituting into the first equation we get $-x + 2(x + 5) = 8$

$$x + 10 = 8$$

$$x = -2$$

Substituting into the second equation we get $2 + 2y = 8$

$$\therefore y = 3$$

$$(x, y) = (-2, 3)$$

2. Find b if a is constant such that $3ax - 8b = 4x + 6a$ for all values of x . Since the equation is true for all values of x it means that x can be set to 0. Thus

$$-8b = 6a$$

It also means that $x = 1$. $\therefore 3a - 8b = 4 + 6a$. Since $-8b = 6a$

$$3a + 6a = 4 + 6a$$

$$3a = 4 \therefore a = \frac{4}{3}$$

Now that we have a we are able to find b .

$$-8b = 6a$$

$$-8b = 8 \therefore b = -1$$

3. Find all ordered pairs (x, y) such that $x + y = 6$ and $x^2 + xy + 2y^2 = 32$. From the first equation we get that $x = 6 - y$. By substitution into the second equation we get

$$(6 - y)^2 + y(6 - y) + 2y^2 = 32$$

$$36 - 12y + y^2 + 6y - y^2 + 2y^2 = 32$$

$$2y^2 - 6y + 4 = 0$$

$$y^2 - 3y + 2 = 0$$

$$(y - 1)(y - 2) = 0$$

$\therefore y = 1, 2$ Then substitute y into the first equation $x + y = 6$ To get that $x = 5, 4$. Therefore the ordered pairs are $(4, 2), (5, 1)$.

4. Solve the system of equations:

$$x + 2y - z = 5$$

$$3x + 2y + z = 11$$

$$(x + 2y)^2 - z^2 = 15$$

Here I can identify the $x + 2y$ in equation 1 and 3. $x + 2y = 5 + z$

$$(5 + z)^2 - z^2 = 15 \quad (1)$$

$$25 + 10z + z^2 - z^2 = 15 \quad (2)$$

$$10z + 25 = 15 \quad (3)$$

$$10z = -10 \quad (4)$$

$$z = -1 \quad (5)$$

Now that we have z we can substitute it into the other equations.

$$x + 2y + 1 = 5 \quad x + 2y = 4 \quad (6)$$

$$3x + 2y - 1 = 11 \quad 3x + 2y = 12 \quad (7)$$

Now the last equation can be rewritten as $2x + (x + 2y) = 12$ $2x + 4 = 12 \therefore x = 4$ Now we substitute x back into the first equation to get $4 + 2y = 4 \therefore y = 0$.

$$(x, y, z) = (4, 0, -1)$$

2 Elimination Exercises

1. Find all ordered pairs (a, b) such that $3a - 2b = -8$ and $5a + 4b = 5$

Multiply the first equation by 2 so that $-2b$ becomes $-4b$, resulting in an easy addition of the two equations.

$$6a - 4b = -16$$

Now add the two equations and we get

$$11a = -11$$

Thus $a = -1$. Substitute back into the first equation to get

$$-3 - 2b = -8 \therefore -2b = -5, b = \frac{5}{2}$$

Thus the ordered pair is $(a, b) = (-1, \frac{5}{2})$.

2. Let x and y be real numbers satisfying $\frac{2}{x} = \frac{y}{3} = \frac{x}{y}$

By cross multiplication between the first and second term we get $6 = xy$, second and third terms result in $y^2 = 3x$
And the first and third terms when cross multiplied result in $2y = x^2$ multiplying both sides by x we get $2xy = x^3$

Since we know that $2xy = 6$ This means that $x^3 = 2(6)$, $x^3 = 12$

3. Find all order pairs (x, y) such that $x^3y^2 = 2^{34}$ and $x^4y = 2^{32}$ Square both sides in the second equation so that we can get y^2 , $x^8y^2 = 2^{64}$ Now we divide the second equation by the first

$$\frac{x^8y^2}{x^3y^2} = \frac{2^{64}}{2^{34}}$$

$$x^5 = 2^{30} \therefore x = 6$$

Substituting this into the first equation we get

$$64^3(y^2) = 2^{34} \therefore y = 256$$

$$(x, y) = (64, 256)$$

4. Find the solution to the system of equations

$$\begin{cases} x + 2z = 1 \\ -xy + 2z = 13 \\ xy + z^2 = 14 \end{cases}$$

I can directly see a similarity between the second and the last equation. Adding the two equations together we can cancel out the xy . This results in

$$\begin{aligned} z^2 + 2z &= -1 \\ z^2 + 2z + 1 &= 0 \\ (z + 1)^2 &= 0 \therefore z = -1 \end{aligned}$$

Substituting z into the first equation we get $x - 2 = 1 \therefore x = 3$ Substitute x and z into equation two resulting in $-3y - 2 = 13$

$$-3y = 15 \therefore y = -5$$

$$\boxed{(x, y, z) = (3, -5, -1)}$$

5. Find all r and s such that $r^2 - 2rs = 27$ and $rs - 2s^2 = 9$ The equations can be rewritten as $r(r - 2s) = 27$ and $s(r - 2s) = 9$. Divide the equations to get rid of the factors

$$\begin{aligned} \frac{r(r - 2s)}{s(r - 2s)} &= \frac{27}{9} \\ \frac{r}{s} &= 3 \therefore r = 3s \end{aligned}$$

Substitute into the second equation to get $3s^2 - 2s^2 = 9$

$$s^2 = 9 \therefore s = \pm 3$$

When $s = 3, r = 9$ and when $s = -3, r = -9$

$$\boxed{(r, s) = (3, 9), (-3, -9)}$$

3 Larger Systems of Linear Equations

1. Solve each of the following systems of equations